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Para Inf.

Defⁿ: $X = (X_1, \dots, X_n)$ has pmf/pdf $f(x|\theta) : \theta \in \Theta$
A statistic $T(X)$ is called complete for θ if
for any fn. g s.t. $E_\theta(g(T)) = 0 \quad \forall \theta \in \Theta \Rightarrow$
 $P_\theta(g(T) = 0) = 1 \quad \forall \theta \in \Theta$

Ex. 1) $X_1, \dots, X_n \sim \text{Ber}(\theta) : T(X) = \sum_{i=1}^n X_i$ is complete for θ
(& sufficient)

2) $X_1, \dots, X_n \sim \text{Poi}(\lambda) : T(X) = \sum_{i=1}^n X_i \sim \text{Poi}(n\lambda)$

Consider
any g

$$E_\lambda[g(T)] = 0 \quad \forall \lambda$$

$$\Rightarrow \sum_{t=0}^{\infty} g(t) \frac{e^{-n\lambda} (n\lambda)^t}{t!} = 0 \quad \forall \lambda > 0$$

$$\Rightarrow \sum_{t=0}^{\infty} g(t) \frac{n^t \lambda^t}{t!} = 0 \quad \forall \lambda > 0$$

power series in λ

$$\Rightarrow g(t) n^t = 0 \quad \forall t = 0, 1, 2, \dots$$

$$\Rightarrow g(t) = 0 \quad \forall t = 0, 1, 2, \dots$$

$$P_\lambda[g(T) = 0] = P[t \in \{0, 1, \dots\}] = 1$$

3) $X_1, \dots, X_n \sim U(0, \theta) \quad \theta > 0$ Claim $T(X) = X_{(n)}$ is complete

$$f_{X_{(n)}}(x) = \frac{nx^{n-1}}{\theta^n} \mathbb{1}_{(0 < x < \theta)}$$

Consider any $g(\cdot)$ s.t.

$$\mathbb{E}_\theta [g(X_{(n)})] = 0 \quad \forall \theta > 0$$

$$\int_0^\theta g(x) \frac{n x^{n-1}}{\theta^n} dx = 0 \quad \forall \theta > 0$$

$$\int_0^\theta g(x) x^{n-1} dx = 0 \quad \forall \theta > 0$$

If g is cont., Leibnitz Integral Rule

$$\left[\begin{array}{l} f(x, \theta) \text{ cont. w.r.t. } x \text{ \& } \theta \\ a(\theta), b(\theta) \text{ have cont. derivatives.} \\ \frac{d}{d\theta} \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx = b'(\theta) f(b(\theta), \theta) - a'(\theta) f(a(\theta), \theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial}{\partial \theta} f(x, \theta) dx \\ \text{If } a(\theta) \text{ and } b(\theta) \text{ are constant/free of } \theta \\ \text{then } \frac{d}{d\theta} \int_a^b f(x, \theta) dx = \int_a^b \frac{\partial}{\partial \theta} f(x, \theta) dx \end{array} \right]$$

$$\Rightarrow g(\theta) \theta^{n-1} \cdot 1 = 0 \quad \forall \theta > 0$$

$$\Rightarrow g(x) = 0 \quad \forall x > 0$$

$$\Rightarrow \mathbb{P}_\theta(g(X_{(n)})=0) = \mathbb{P}_\theta(0 < X_{(n)} < \theta) = 1$$

$$g(x) = g^+(x) - g^-(x)$$

$$g^+(x) = \max\{g(x), 0\}$$

$$g^-(x) = \max\{-g(x), 0\}$$

$$\therefore \int_0^\theta g^+(x) x^{n-1} dx = \int_0^\theta g^-(x) x^{n-1} dx \quad \forall \theta > 0$$

$$\mu^+(A) := \int_A g^+(x) x^{n-1} dx$$

$$\mu^-(A) := \int_A g^-(x) x^{n-1} dx$$

$$\mu^+(A) = \mu^-(A) \quad \forall A \text{ Borel on } (0, \infty)$$

$$g^+(x) x^{n-1} = g^-(x) x^{n-1} \text{ for } \mu \text{ a.e.}$$

Q: Does completeness imply sufficiency?

$\Rightarrow$ No

Ex $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1) \quad n \geq 2$

• Show that $T(X) = \frac{X_1 + X_2}{2} \sim N(\theta, 1/2)$ is complete

• But $T(X)$ is not sufficient for θ .

Consider a real fn. g $E_\theta(g(T)) = 0 \quad \forall \theta \in \mathbb{R}$

$$\Rightarrow \int g\left(\frac{t}{2}\right) \frac{1}{\sqrt{\frac{1}{2} \cdot 2\pi}} e^{-\frac{1}{2 \cdot \frac{1}{2}}(t-\theta)^2} dt = 0$$

$$\Rightarrow \int_{-\infty}^{\infty} g(t) e^{-(t^2 - 2t\theta + \theta^2)} dt = 0$$

$$\Rightarrow \int_{-\infty}^{\infty} \underbrace{g(t) e^{-t^2}}_{f(t)} e^{2t\theta} dt = 0$$

Let $f(t)$ be some real-valued fn. Unilateral Laplace transform of f is

$$F(s) := \int_0^{\infty} e^{-st} f(t) dt, \quad s \in \mathbb{R} \text{ or } \mathbb{C} \text{ s.t.}$$

the $\lim_{a \rightarrow \infty} \int_0^a e^{-st} f(t) dt$ exists.

Bilateral Laplace transform

$$F(s) := \int_{-\infty}^{\infty} e^{-st} f(t) dt$$

If f and g are two real fns. s.t. their Laplace transforms (unilateral or bilateral)

$$F(s) = \int_{-\infty}^{\infty} e^{-st} f(t) dt$$

$$G(s) = \int_{-\infty}^{\infty} e^{-st} g(t) dt \quad \forall s > k > 0 \text{ for some } k$$

$\Rightarrow$ uniqueness of Laplace transform

$$f(t) = g(t) \quad \mu \text{ a.e. (Lebesgue)}$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-st} f(t) dt = 0 \quad \forall s \in \mathbb{R} \quad \text{where } \left. \begin{array}{l} f(t) = g(t) e^{-t^2} \\ \& s = -2\theta \end{array} \right\}$$

$$\Rightarrow f(t) = 0 \quad \mu \text{ a.e.}$$

$$\Rightarrow g(t) = 0 \quad \mu \text{ a.e.}$$

$$\mathbb{P}_\theta[g(T) = 0] = 1 \quad \forall \theta$$

$$T = \frac{X_1 + X_2}{2} \sim N(\theta, 1/2) \text{ complete (not suff.)}$$

Thm. If $T(\underline{X})$ is complete and sufficient for θ , then $T(\underline{X})$ is also minimal sufficient for θ [Pf: due]. But converse not true.

Minimal Suff. $\nRightarrow$ Complete

Ex. 1) $X_1, \dots, X_n$ iid $N(\theta, \theta^2)$. $\theta \in \mathbb{R}$

$T(\underline{X}) = (\bar{X}, S^2)$ is minimal suff for θ

$$\text{Know } \mathbb{E} S^2 = \theta^2$$

$$\mathbb{E} \bar{X}^2 = \theta^2 + \frac{\theta^2}{n} = \theta^2 \left(\frac{n+1}{n} \right)$$

$$\mathbb{E} \left(\frac{n}{n+1} \bar{X}^2 \right) = \theta^2$$

$$\text{Take } g(T) = S^2 - \frac{n}{n+1} \bar{X}^2$$

$$\mathbb{E}_\theta[g(T)] = 0 \quad \forall \theta$$

$$\text{But } \mathbb{P}_\theta \left[S^2 - \frac{n \bar{X}^2}{n+1} = 0 \right] = 0 \quad \forall \theta$$

Ex: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} U(\theta, \theta+1)$

Show min suff stat $T(X) = (X_{(1)}, X_{(n)})$

T not complete b/c. Check.

Basu's Theorem

If $S(X)$ is complete and sufficient statistic for θ
c.s.s.

(X has pmf/pdf $f(x|\theta): \theta \in \Theta$) and $A(X)$ is an ancillary statistic for θ then $S(X)$ is independent of $A(X)$

(converse not true)

Pf: Consider $h_a(s) := \mathbb{P}_\theta(A(X) \leq a | S(X) = s)$ does not depend on θ b/c.

S is suff for θ

$$\begin{aligned} \mathbb{P}_\theta(A(X) \leq a) &= \mathbb{E}[\mathbb{1}_{A(X) \leq a}] \\ &= \mathbb{E}[\mathbb{E}\{\mathbb{1}_{A(X) \leq a} | S(X)\}] \\ &= \mathbb{E}[h_a(S(X))] \end{aligned}$$

↓
does not depend on θ
b/c $A(X)$ ancillary for θ

$$\text{We have } \mathbb{P}[A(X) \leq a] = \mathbb{E}[h_a(S(X))]$$

$$\Rightarrow \mathbb{E}_\theta \underbrace{[h_a(S(X)) - \mathbb{P}[A(X) \leq a]]}_{g(S)} = 0 \quad \forall \theta \in \Theta$$

Since S is complete for θ , then $\mathbb{P}[h_a(S(X)) = \mathbb{P}[A(X) \leq a]] = 1 \quad \forall \theta$

$$\Rightarrow h_a(s) = \mathbb{P}[A(X) \leq a | S(X) = s] = \mathbb{P}[A(X) \leq a] \quad \forall s \text{ in the support of } S(X)$$

$$\Rightarrow S(X) \perp\!\!\!\perp A(X)$$

(Application of Basu's Theorem)

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$

Fix $\sigma^2 = \sigma_0^2$ (Known)

Check: $\bar{X}$ is complete and suff for μ
 $\sim N(\mu, \sigma_0^2/n)$

Check: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is ancillary for μ

Basu's Theorem $\Rightarrow \bar{X} \perp\!\!\!\perp S^2 \longrightarrow (*)$

Since $(*)$ holds for all $\sigma_0 \in \mathbb{R}^+$, then it holds for all σ

Ex. $X \sim N(0, \sigma^2 I_n)$

Claim: $\|X\| \perp\!\!\!\perp \frac{X}{\|X\|}$

$\|X\|^2 = \sum_i X_i^2$ complete suff for σ^2

$\frac{X}{\|X\|}$ is ancillary for σ^2

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda) \longrightarrow \text{scale}$

$f(x|\lambda) = \frac{1}{\lambda} e^{-x/\lambda} \quad \lambda > 0, x > 0$

$\sum X_i$ complete, sufficient, $\frac{X_1}{\sum X_i}$ ancillary

$\therefore \sum X_i \perp\!\!\!\perp \frac{X_1}{\sum X_i}$

$$E X_1 = E \left(\frac{X_1}{\sum X_i} \sum X_i \right) = E \left(\frac{X_1}{\sum X_i} \right) E \left(\sum X_i \right)$$

$$E \left(\frac{X_1}{\sum X_i} \right) = \frac{E X_1}{E \left(\sum X_i \right)} = \frac{\lambda}{n\lambda} = \frac{1}{n}$$